ASSESSMENT TOOL 2

COURSE NAME: MOBILE COMPUTING

COURSE CODE: ITA03

COURSE OUTCOME COVERED

CO2: Compare mobile communication technologies, including multiple access techniques and network evolution, based on performance and applications. (BL-1, BL-2, BL-3)

QUESTIONS: 5 **Total Marks: 100**

Annexure A

EXPERIMENTATION

Code of Conduct:

I **J. SAI DHEERAJ (192311191)** (Name / Reg No) certify that this submission is my original work

and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: **J. SAI DHEERAJ**

Annexure A - QUESTIONS

1) Figma-Based Mobile IP Communication Workflow Analysis (BL4 – Analyze)

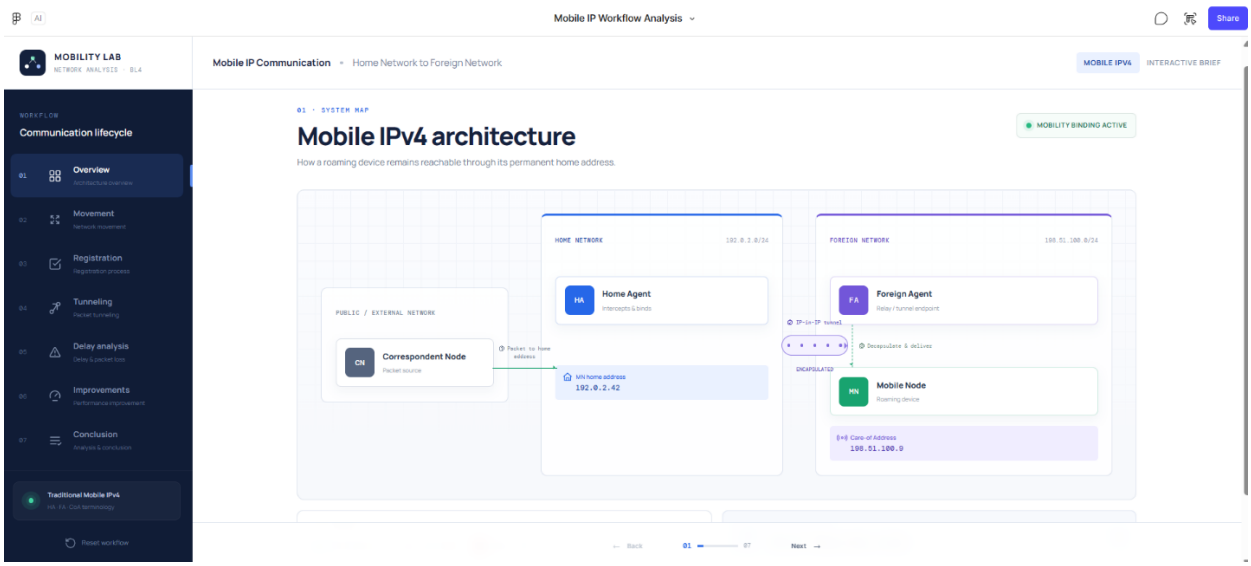
Design an interactive Figma prototype illustrating the complete workflow of Mobile IP communication when a Mobile Node moves from its Home Network to a Foreign Network. Your design should include the Home Agent (HA), Foreign Agent (FA), Correspondent Node (CN), Care-of Address (CoA), Registration Process, and Packet Tunneling.

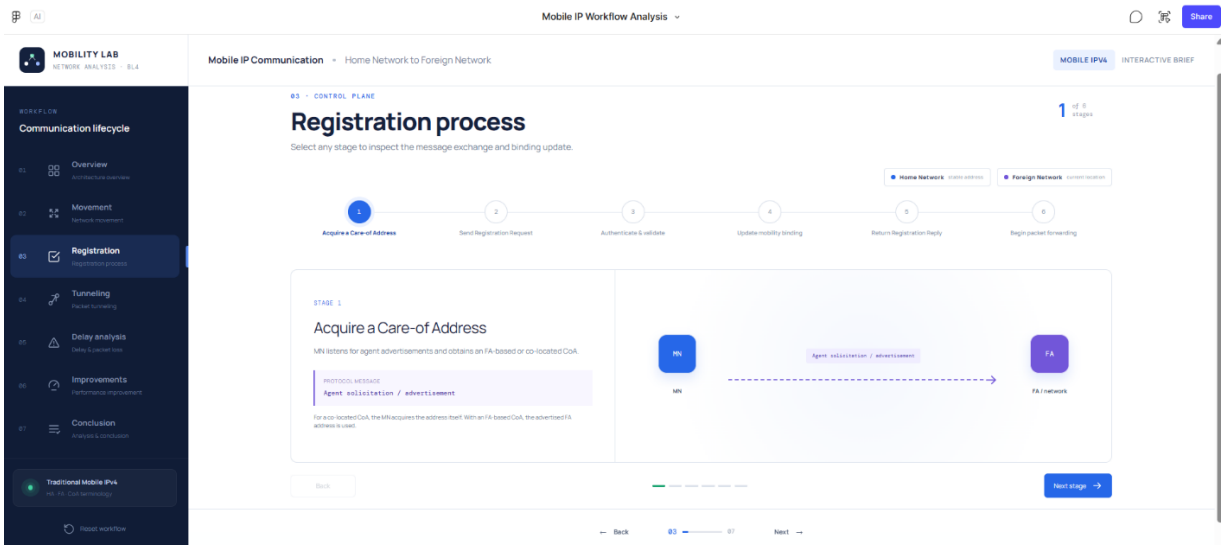
Analyze the communication process by identifying:

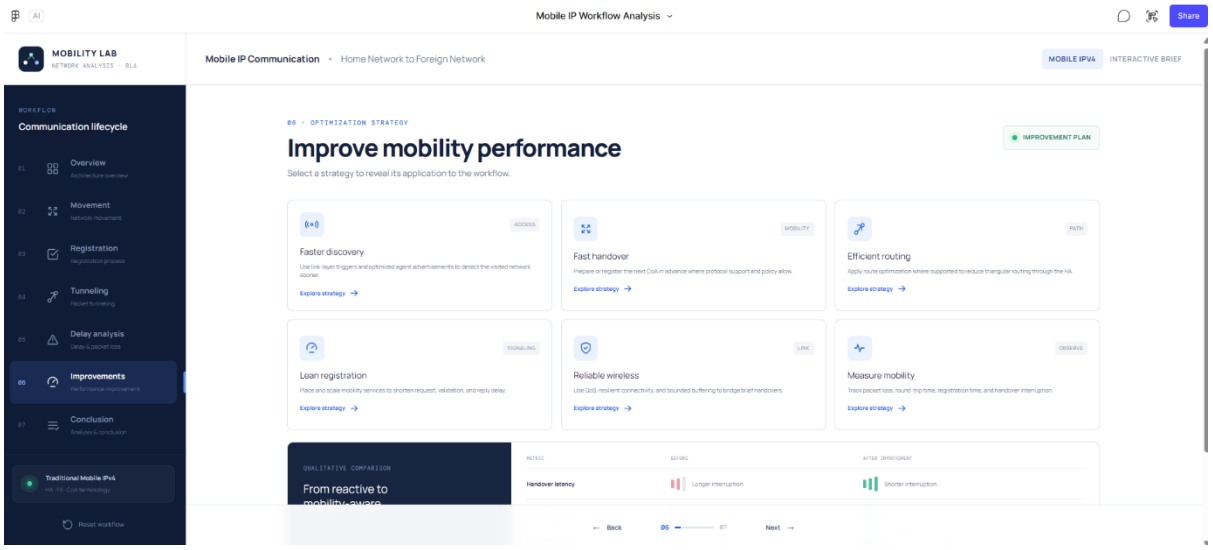
- The purpose of each network component.
- Points where packet delay or packet loss may occur.
- The effect of node mobility on communication performance.
- Possible improvements to reduce latency.

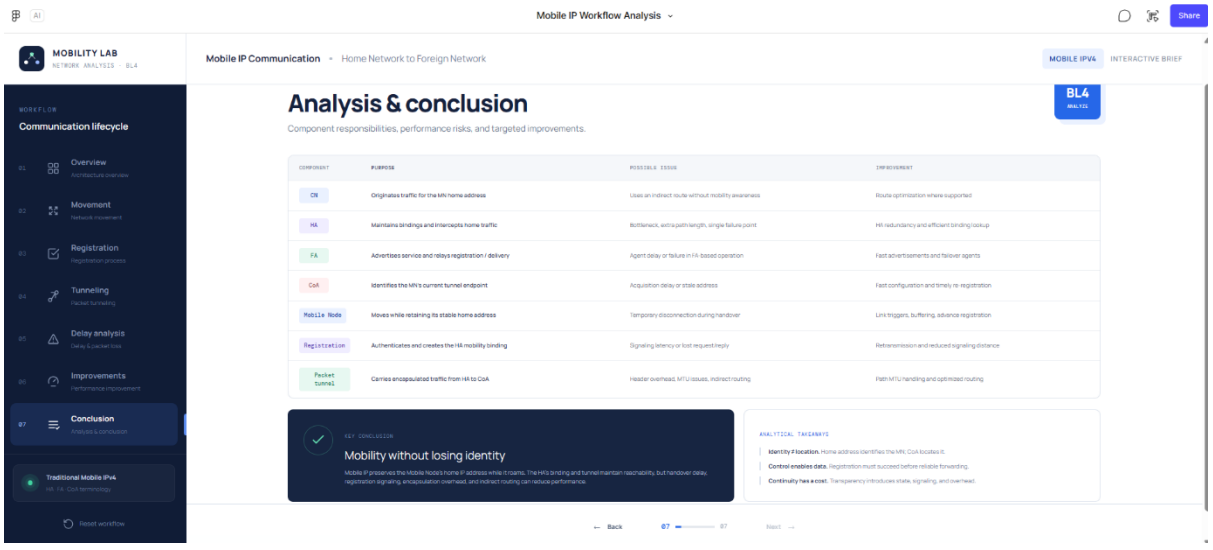
Deliverables:

- Figma (.fig) design
- GitHub repository containing exported images/PDF and documentation (README).




Mobile IP Workflow Analysis
Share
MOBILITY LAB
NETWORK ANALYSIS · BL4
Mobile IP Communication · Home Network to Foreign Network
MOBILE IPV4
INTERACTIVE BRIEF
WORKFLOW
Communication lifecycle
Overview
Movement
Registration
Tunneling
Delay analysis
Improvements
Conclusion
Traditional Mobile IPv4
Reset workflow
03 · CONTROL PLANE
Registration process
Select any stage to inspect the message exchange and binding update.
1 of 6 stages
Home Network
Foreign Network
1 Acquire a Care-of Address
2 Send Registration Request
3 Authenticate & validate
4 Update mobility binding
5 Return Registration Reply
6 Begin packet forwarding
STAGE 1
Acquire a Care-of Address
MN listens for agent advertisements and obtains an FA-based or co-located CoA.
PROTOCOL MESSAGE
Agent solicitation / advertisement
For a co-located CoA, the MN acquires the address itself. With an FA-based CoA, the advertised FA address is used.
MN
Agent solicitation / advertisement
FA
MN
FA / network
Back
Next stage
← Back
03
07
Next →


Mobile IP Workflow Analysis
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NETWORK ANALYSIS · BL4
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06 · OPTIMIZATION STRATEGY
Improve mobility performance
Select a strategy to reveal its application to the workflow.
IMPROVEMENT PLAN
ACCESS
Faster discovery
Use link-layer triggers and optimized agent advertisements to detect the visited network sooner.
Explore strategy →
MOBILITY
Fast handover
Prepare or register the next CoA in advance where protocol support and policy allow.
Explore strategy →
PATH
Efficient routing
Apply route optimization where supported to reduce triangular routing through the HA.
Explore strategy →
SIGNALING
Lean registration
Place and scale mobility services to shorten request, validation, and reply delay.
Explore strategy →
LINK
Reliable wireless
Use QoS, resilient connectivity, and bounded buffering to bridge brief handovers.
Explore strategy →
OBSERVE
Measure mobility
Track packet loss, round-trip time, registration time, and handover interruption.
Explore strategy →
QUALITATIVE COMPARISON
From reactive to
METRIC
BEFORE
AFTER IMPROVEMENT
Handover latency
Longer interruption
Shorter interruption
← Back
06
07
Next →


Mobile IP Workflow Analysis
Share
MOBILITY LAB
NETWORK ANALYSIS · BL4
Mobile IP Communication · Home Network to Foreign Network
MOBILE IPV4
INTERACTIVE BRIEF
WORKFLOW
Communication lifecycle
Overview
Movement
Registration
Tunneling
Delay analysis
Improvements
Conclusion
Traditional Mobile IPv4
Reset workflow
Analysis & conclusion
Component responsibilities, performance risks, and targeted improvements.
BL4
COMPONENT
PURPOSE
POSSIBLE ISSUE
IMPROVEMENT
CN
Originates traffic for the MN home address
Uses an indirect route without mobility awareness
Route optimization where supported
HA
Maintains bindings and intercepts home traffic
Bottleneck, extra path length, single failure point
HA redundancy and efficient binding lookup
FA
Advertises service and relays registration / delivery
Agent delay or failure in FA-based operation
Fast advertisements and failover agents
CoA
Identifies the MN's current tunnel endpoint
Acquisition delay or stale address
Fast configuration and timely re-registration
Mobile Node
Moves while retaining its stable home address
Temporary disconnection during handover
Link triggers, buffering, advance registration
Registration
Authenticates and creates the HA mobility binding
Signaling latency or lost request/reply
Retransmission and reduced signaling distance
Packet tunnel
Carries encapsulated traffic from HA to CoA
Header overhead, MTU issues, indirect routing
Path MTU handling and optimized routing
KEY CONCLUSION
Mobility without losing identity
Mobile IP preserves the Mobile Node's home IP address while it roams. The HA's binding and tunnel maintain reachability, but handover delay, registration signaling, encapsulation overhead, and indirect routing can reduce performance.
ANALYTICAL TAKEAWAYS
Identity ≠ location. Home address identifies the MN; CoA locates it.
Control enables data. Registration must succeed before reliable forwarding.
Continuity has a cost. Transparency introduces state, signaling, and overhead.
← Back
07
07
Next →

2) Comparative Network Architecture Design Using Figma (BL4 – Analyze)

Using Figma or draw.io, create a comparative architectural diagram showing both a Cellular Wireless Network and a Mobile Ad Hoc Network (MANET) operating in the same smart city environment.

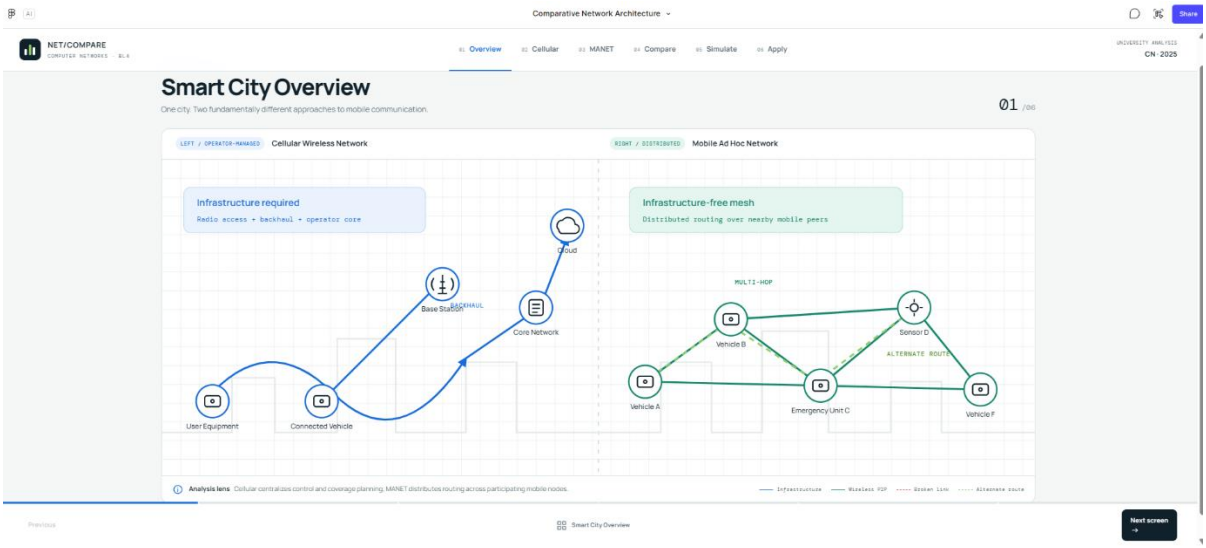
Analyze the architectures by comparing:

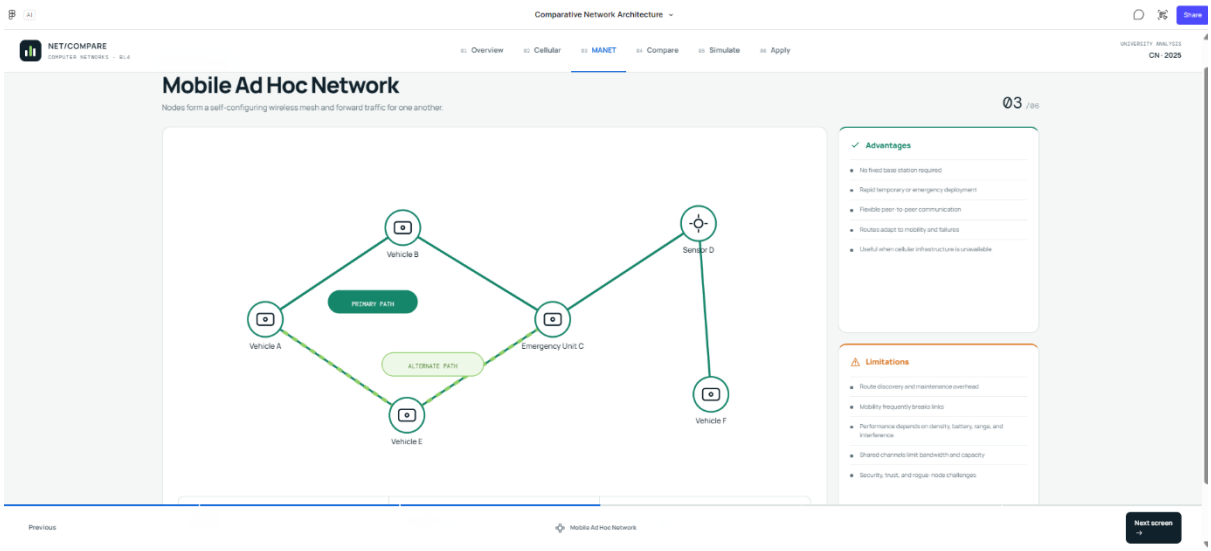
- Infrastructure dependency.
- Routing mechanisms.
- Scalability.
- Fault tolerance.
- Mobility support.
- Suitable application scenarios.

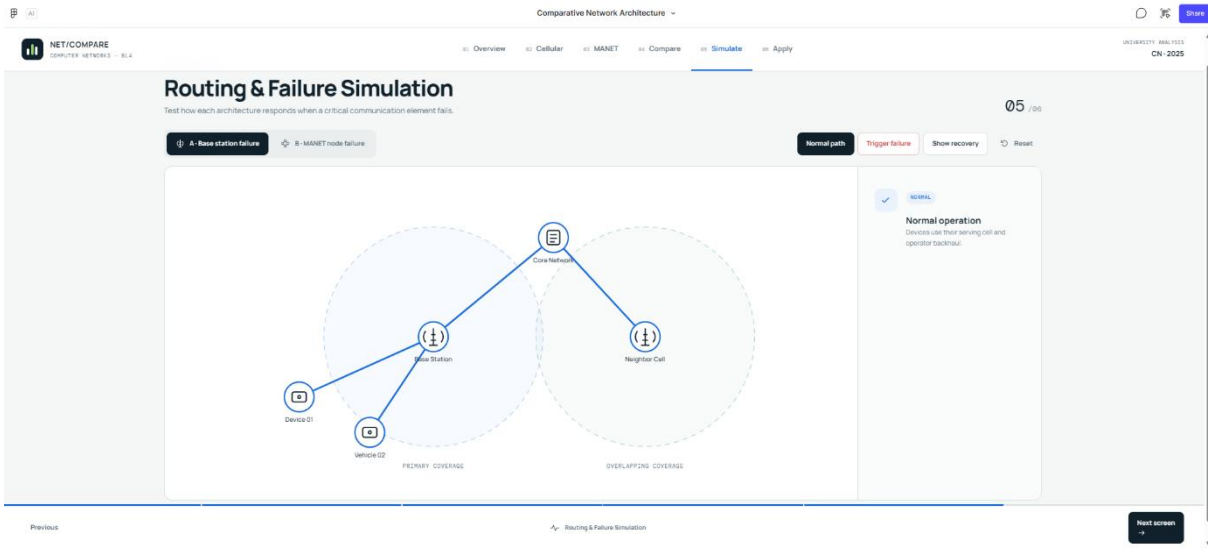
Highlight the advantages and limitations of each architecture through annotations in your design.

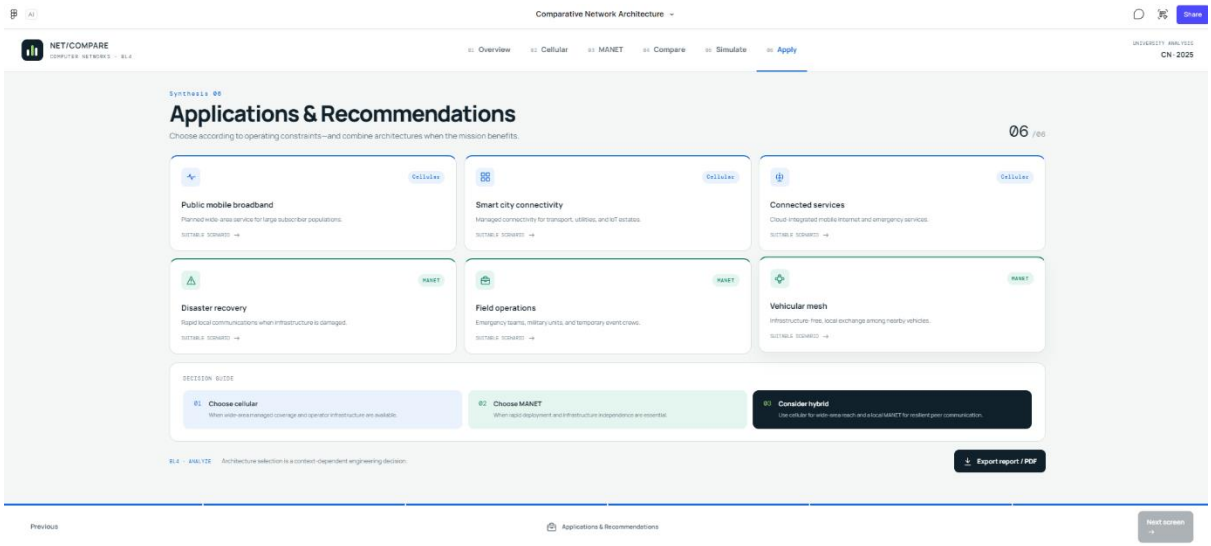
Deliverables:

- Editable design file.
- GitHub repository with screenshots, comparison report, and design assets.




Comparative Network Architecture
Share
NET/COMPARE
COMPUTER NETWORKS · BL4
01 Overview
02 Cellular
03 MANET
04 Compare
05 Simulate
06 Apply
UNIVERSITY ANALYSIS
CN · 2025
Mobile Ad Hoc Network
Nodes form a self-configuring wireless mesh and forward traffic for one another.
03 /06
Vehicle B
Sensor D
PRIMARY PATH
Vehicle A
Emergency Unit C
ALTERNATE PATH
Vehicle E
Vehicle F
Advantages
No fixed base station required
Rapid temporary or emergency deployment
Flexible peer-to-peer communication
Routes adapt to mobility and failures
Useful when cellular infrastructure is unavailable
Limitations
Route discovery and maintenance overhead
Mobility frequently breaks links
Performance depends on density, battery, range, and interference
Shared channels limit bandwidth and capacity
Security, trust, and rogue-node challenges
Previous
Mobile Ad Hoc Network
Next screen


Comparative Network Architecture
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NET/COMPARE
COMPUTER NETWORKS · BL4
01 Overview
02 Cellular
03 MANET
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05 Simulate
06 Apply
UNIVERSITY ANALYSIS
CN · 2025
Routing & Failure Simulation
Test how each architecture responds when a critical communication element fails.
05 /06
A - Base station failure
B - MANET node failure
Normal path
Trigger failure
Show recovery
Reset
Core Network
Base Station
Neighbor Cell
Device 01
Vehicle 02
PRIMARY COVERAGE
OVERLAPPING COVERAGE
NORMAL
Normal operation
Devices use their serving cell and operator backhaul.
Previous
Routing & Failure Simulation
Next screen


Comparative Network Architecture
Share
NET/COMPARE
COMPUTER NETWORKS · BL4
01 Overview
02 Cellular
03 MANET
04 Compare
05 Simulate
06 Apply
UNIVERSITY ANALYSIS
CN · 2025
Synthesis 06
Applications & Recommendations
Choose according to operating constraints—and combine architectures when the mission benefits.
06 /06
Cellular
Public mobile broadband
Planned wide-area service for large subscriber populations.
SUITABLE SCENARIO
Cellular
Smart city connectivity
Managed connectivity for transport, utilities, and IoT estates.
SUITABLE SCENARIO
Cellular
Connected services
Cloud-integrated mobile Internet and emergency services.
SUITABLE SCENARIO
MANET
Disaster recovery
Rapid local communications when infrastructure is damaged.
SUITABLE SCENARIO
MANET
Field operations
Emergency teams, military units, and temporary event crews.
SUITABLE SCENARIO
MANET
Vehicular mesh
Infrastructure-free, local exchange among nearby vehicles.
SUITABLE SCENARIO
DECISION GUIDE
01 Choose cellular
When wide-area managed coverage and operator infrastructure are available.
02 Choose MANET
When rapid deployment and infrastructure independence are essential.
03 Consider hybrid
Use cellular for wide-area reach and a local MANET for resilient peer communication.
BL4 · ANALYZE
Architecture selection is a context-dependent engineering decision.
Export report / PDF
Previous
Applications & Recommendations
Next screen

3) Routing Protocol Visualization and Analysis (BL4 – Analyze)

Develop an interactive visualization using Figma, Mermaid.js, D3.js, or any visualization framework illustrating the operation of one Proactive Routing Protocol and one Reactive

Routing Protocol in a mobile wireless network.

Analyze:

 Route discovery process.

 Route maintenance mechanism.

 Routing overhead.

 Delay in establishing communication.

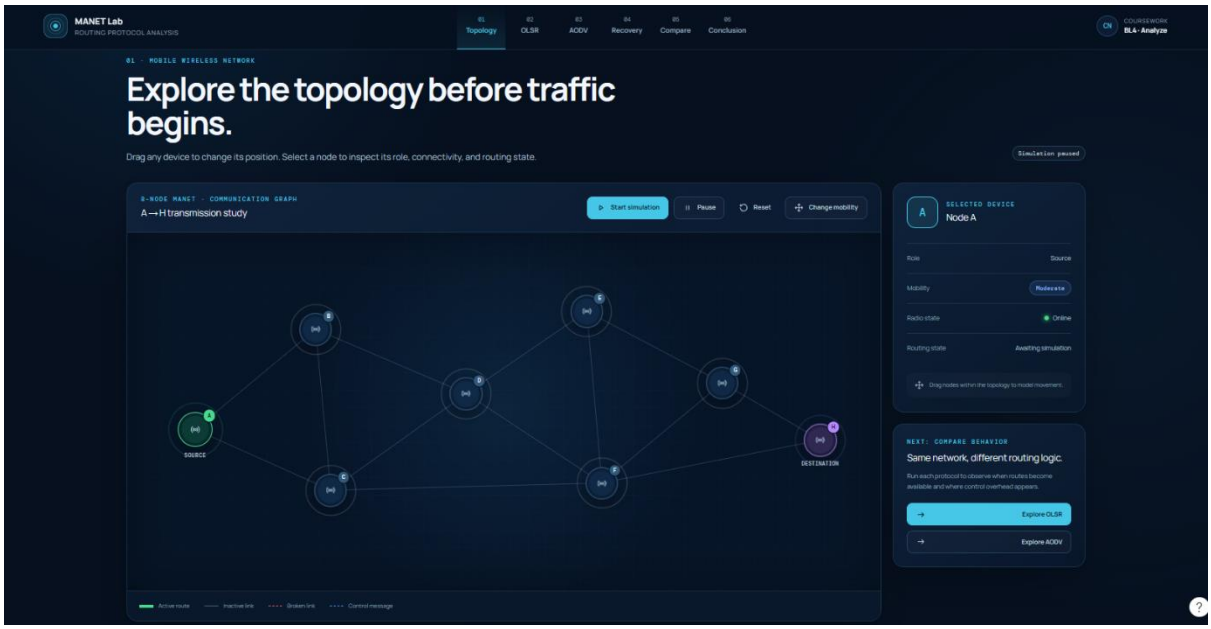
 Performance under high node mobility.

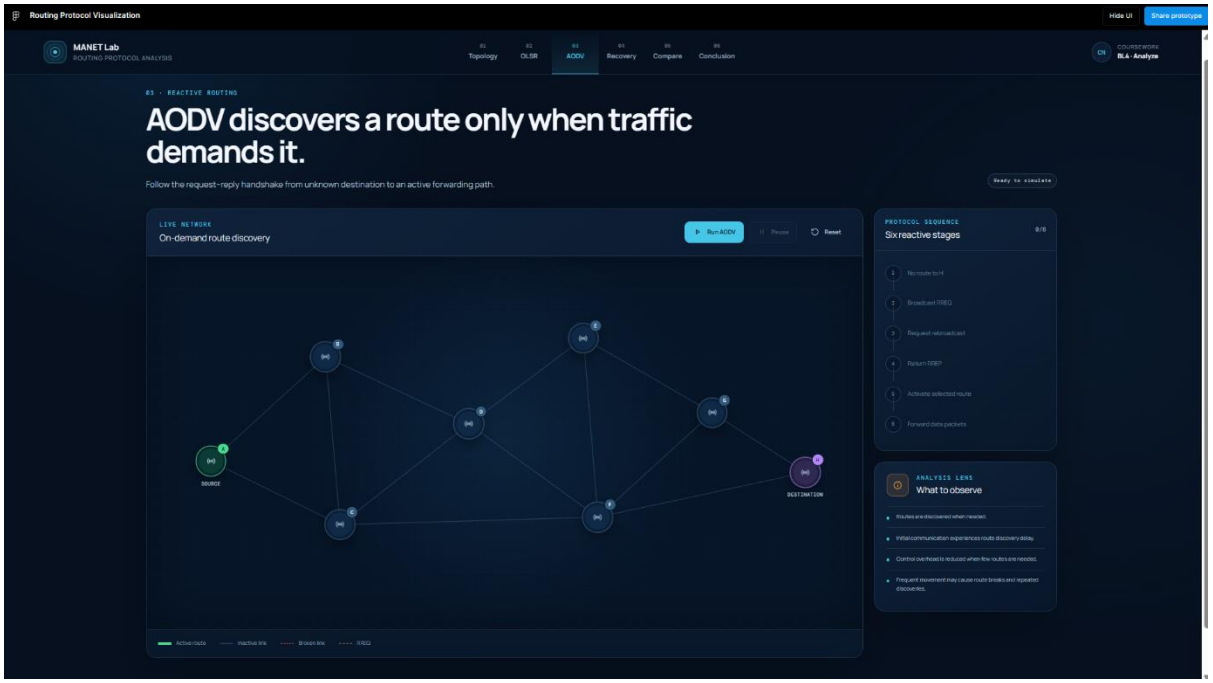
Include a comparative summary explaining which protocol performs better under different mobility conditions.

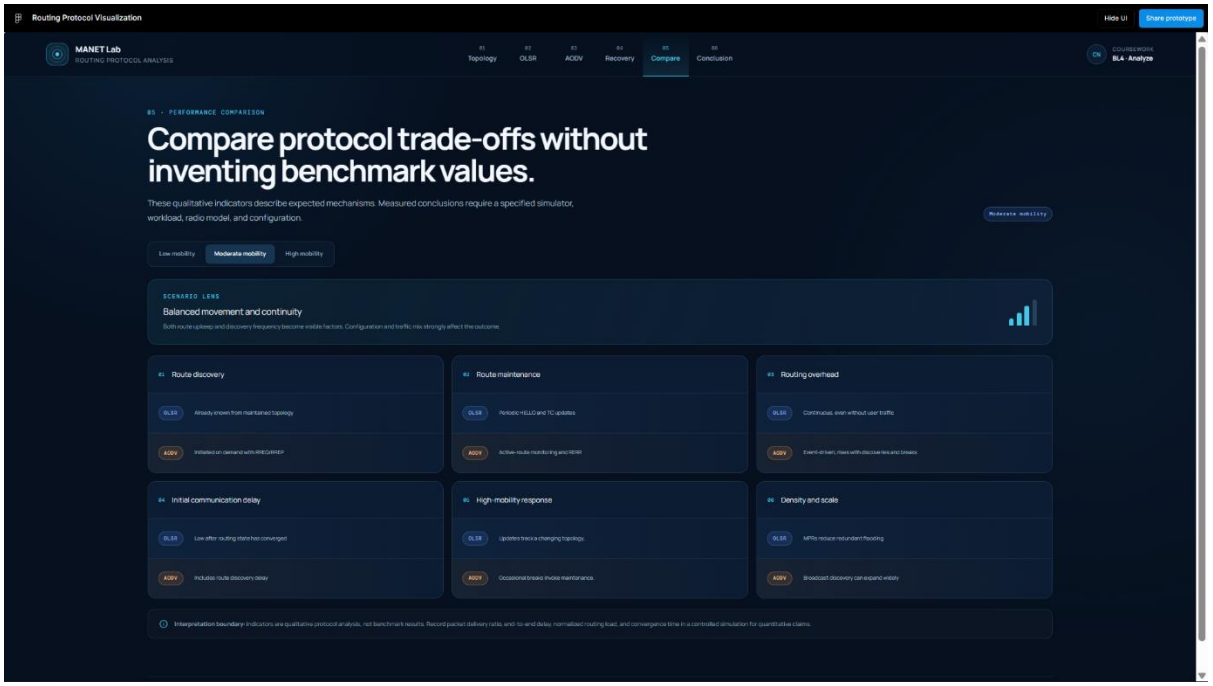
Deliverables:

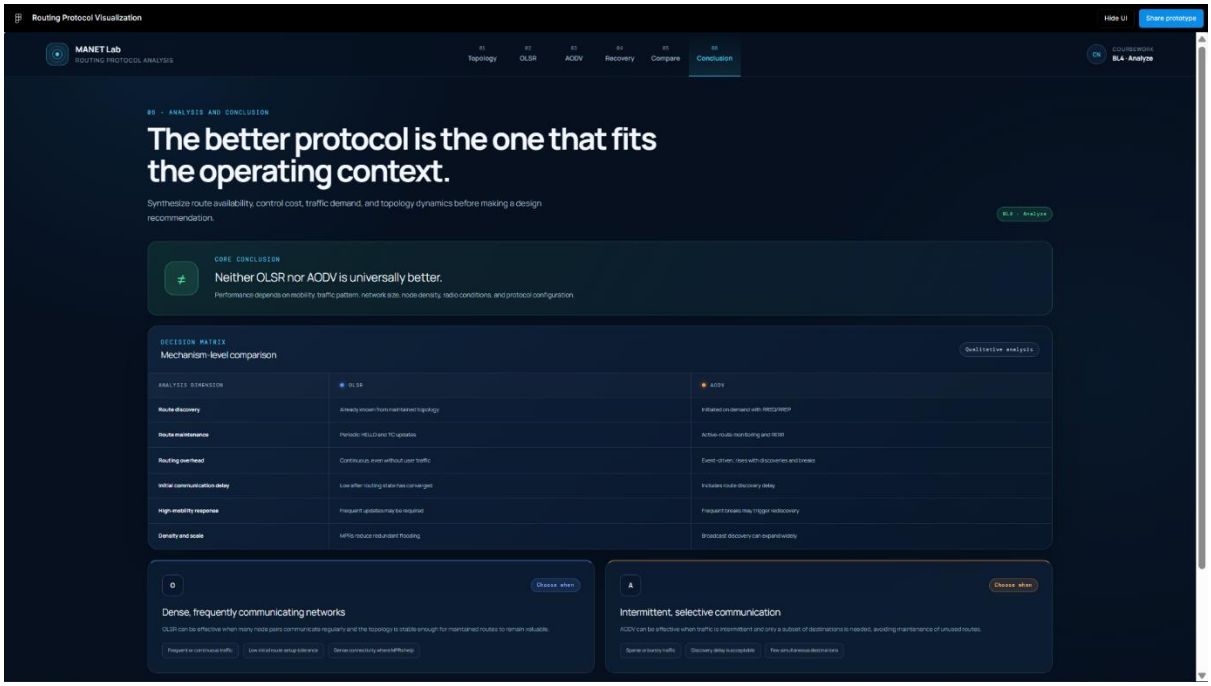
 Source code/design files.

 GitHub repository with implementation, screenshots, and analysis report.




Routing Protocol Visualization
Hide UI
Share prototype
MANET Lab
ROUTING PROTOCOL ANALYSIS
01 Topology
02 OLSR
03 AODV
04 Recovery
05 Compare
06 Conclusion
COURSEWORK
BL4 · Analyze
03 · REACTIVE ROUTING
AODV discovers a route only when traffic demands it.
Follow the request–reply handshake from unknown destination to an active forwarding path.
Ready to simulate
LIVE NETWORK
On-demand route discovery
Run AODV
Pause
Reset
SOURCE
DESTINATION
PROTOCOL SEQUENCE
Six reactive stages
0/6
1 No route to H
2 Broadcast RREQ
3 Request rebroadcast
4 Return RREP
5 Activate selected route
6 Forward data packets
ANALYSIS LENS
What to observe
Routes are discovered when needed.
Initial communication experiences route discovery delay.
Control overhead is reduced when few routes are needed.
Frequent movement may cause route breaks and repeated discoveries.
Active route
Inactive link
Broken link
RREQ


Routing Protocol Visualization
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COURSEWORK
BL4 · Analyze
05 · PERFORMANCE COMPARISON
Compare protocol trade-offs without inventing benchmark values.
These qualitative indicators describe expected mechanisms. Measured conclusions require a specified simulator, workload, radio model, and configuration.
Moderate mobility
Low mobility
Moderate mobility
High mobility
SCENARIO LENS
Balanced movement and continuity
Both route upkeep and discovery frequency become visible factors. Configuration and traffic mix strongly affect the outcome.
01 Route discovery
OLSR Already known from maintained topology
AODV Initiated on demand with RREQ/RREP
02 Route maintenance
OLSR Periodic HELLO and TC updates
AODV Active-route monitoring and RERR
03 Routing overhead
OLSR Continuous, even without user traffic
AODV Event-driven, rises with discoveries and breaks
04 Initial communication delay
OLSR Low after routing state has converged
AODV Includes route discovery delay
05 High-mobility response
OLSR Updates track a changing topology
AODV Occasional breaks invoke maintenance
06 Density and scale
OLSR MPRs reduce redundant flooding
AODV Broadcast discovery can expand widely
Interpretation boundary: Indicators are qualitative protocol analysis, not benchmark results. Record packet delivery ratio, end-to-end delay, normalised routing load, and convergence time in a controlled simulation for quantitative claims.



4) Mobile IP Application Design for Smart Healthcare (BL4 – Analyze)

Design a Smart Healthcare Mobile IP system using Figma that enables doctors, ambulances, hospitals, and patients to remain connected while moving across different wireless networks.

Analyze:

 How Mobile IP maintains uninterrupted communication.

 Challenges during handoff between networks.

 Security vulnerabilities during mobility.

 Possible design improvements to increase reliability.

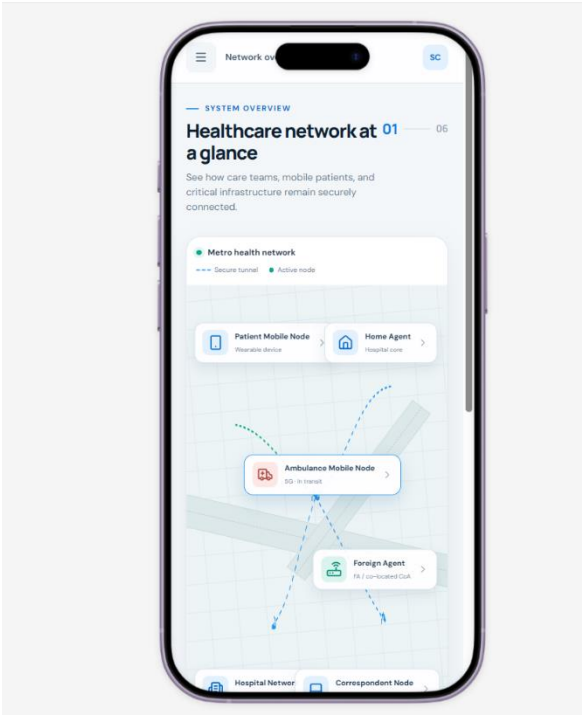
Provide navigation between screens demonstrating different mobility scenarios.

Deliverables:

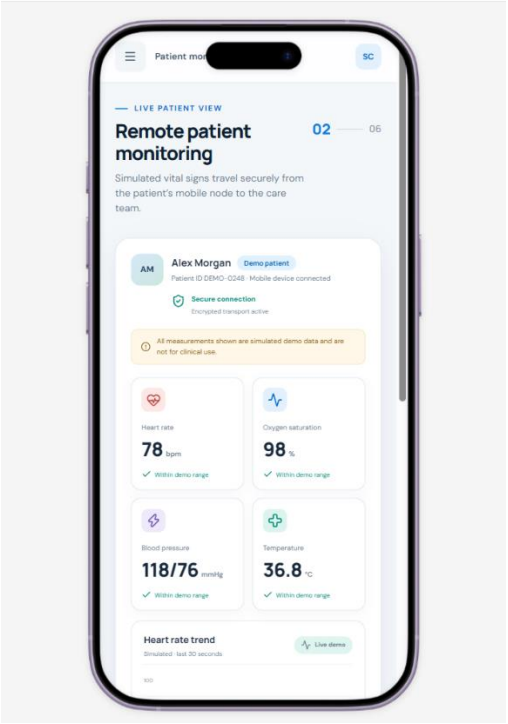
 Interactive prototype.

 GitHub repository containing design files, exported prototype, and documentation.

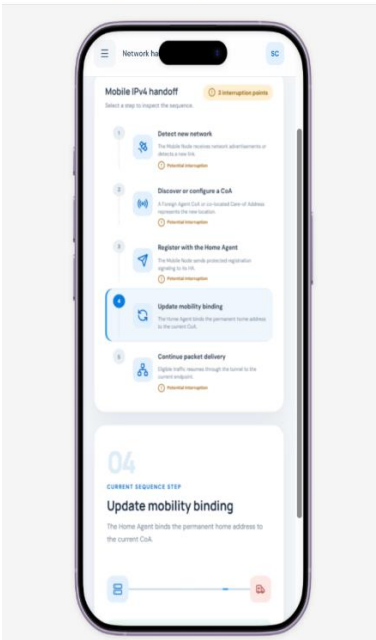
Smart Healthcare Mobility Dashboard



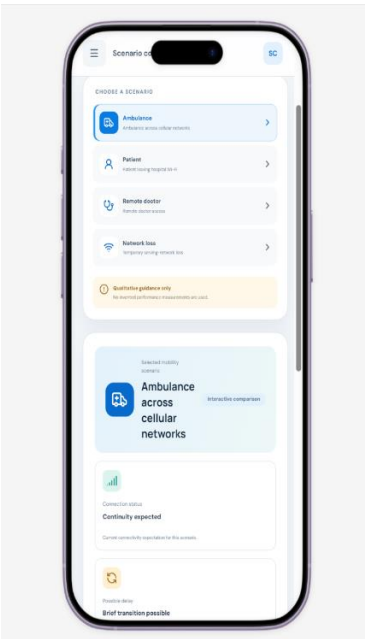
Smart Healthcare Mobility Dashboard



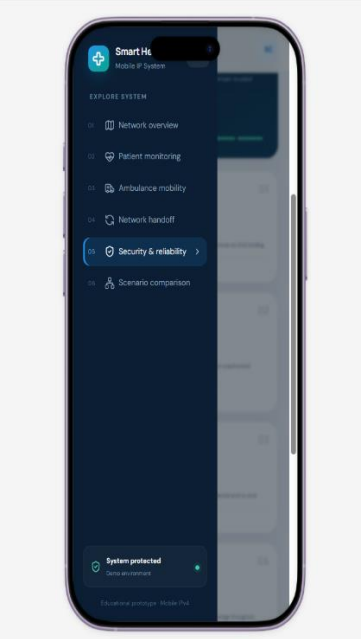
Smart Healthcare Mobility Dashboard



Smart Healthcare Mobility Dashboard



Smart Healthcare Mobility Dashboard



5) Wireless Network Decision Dashboard Using GitHub Project (BL4 – Analyze)

Create a web-based dashboard (using HTML/CSS/JavaScript, React, or another programming framework) or an interactive Figma dashboard that helps network engineers determine whether a given scenario should use Mobile IP, Cellular Networks, or Ad Hoc Networks.

The dashboard should analyze multiple parameters such as:

 Node mobility.

 Infrastructure availability.

 Network size.

 Routing complexity.

 Energy consumption.

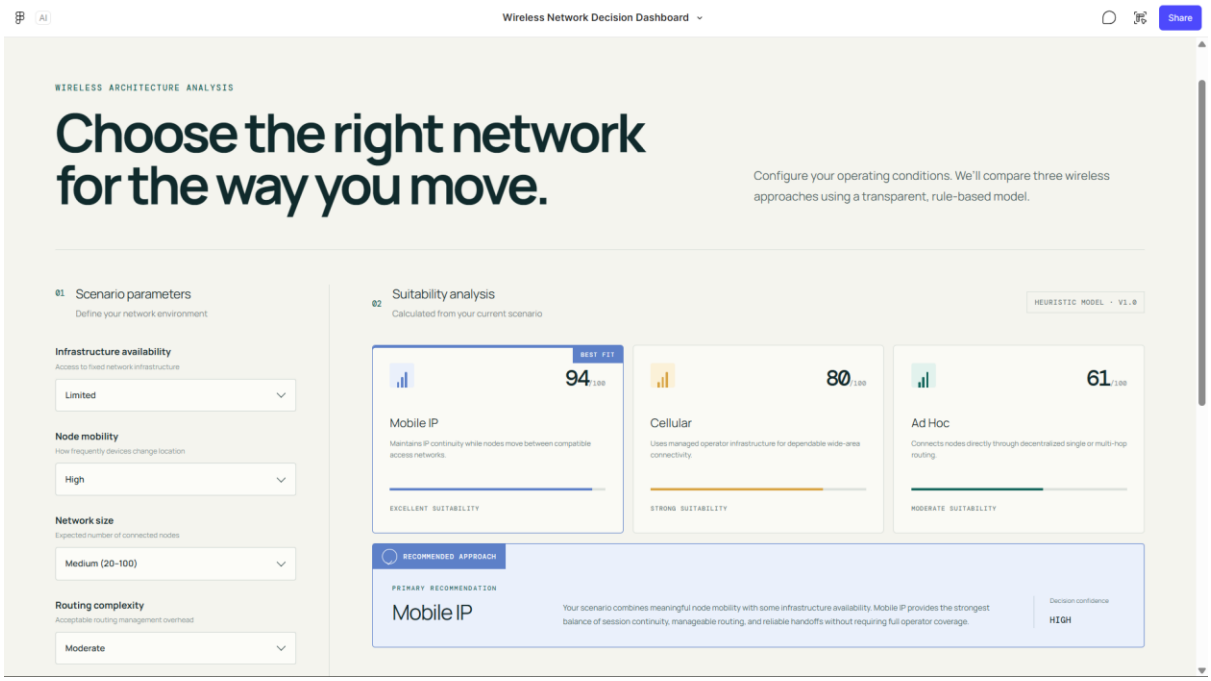
 Reliability requirements.

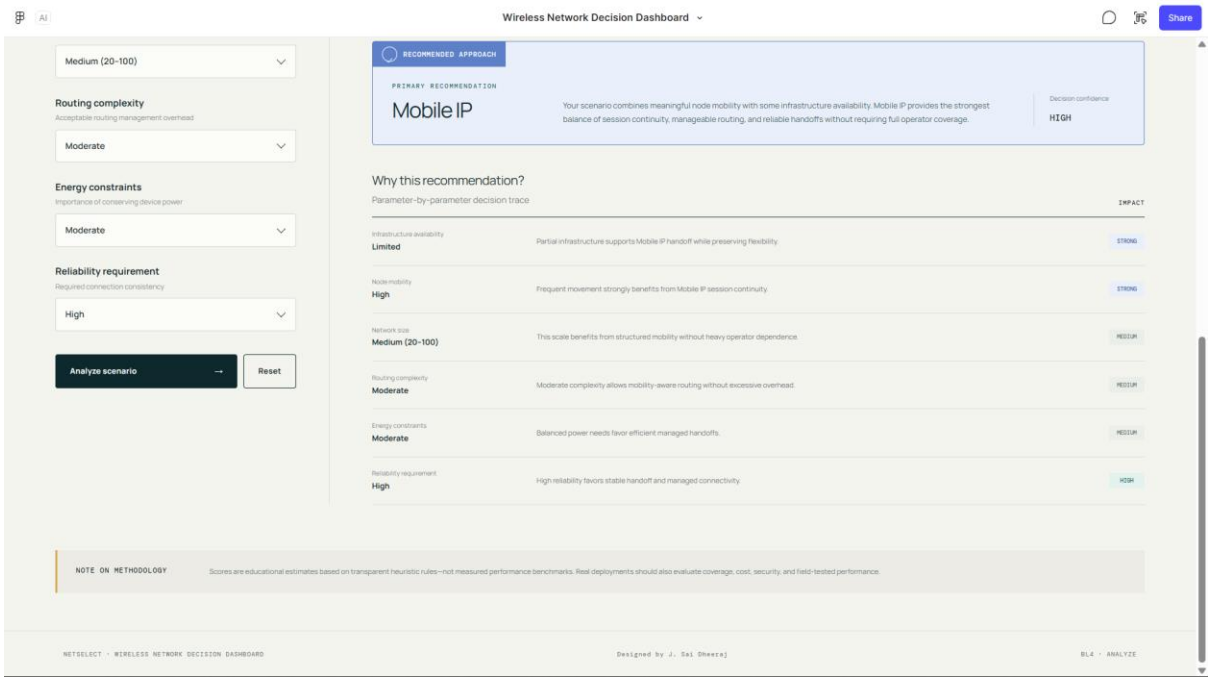
The system should justify its recommendation based on the analyzed parameters.

Deliverables:

 Source code or Figma project.

 GitHub repository with complete code/design, README, screenshots, and analysis report.


Wireless Network Decision Dashboard
Share
WIRELESS ARCHITECTURE ANALYSIS
Choose the right network for the way you move.
Configure your operating conditions. We'll compare three wireless approaches using a transparent, rule-based model.
01 Scenario parameters
Define your network environment
Infrastructure availability
Access to fixed network infrastructure
Limited
Node mobility
How frequently devices change location
High
Network size
Expected number of connected nodes
Medium (20-100)
Routing complexity
Acceptable routing management overhead
Moderate
02 Suitability analysis
Calculated from your current scenario
HEURISTIC MODEL · V1.0
BEST FIT
94/100
Mobile IP
Maintains IP continuity while nodes move between compatible access networks.
EXCELLENT SUITABILITY
80/100
Cellular
Uses managed operator infrastructure for dependable wide-area connectivity.
STRONG SUITABILITY
61/100
Ad Hoc
Connects nodes directly through decentralized single or multi-hop routing.
MODERATE SUITABILITY
RECOMMENDED APPROACH
PRIMARY RECOMMENDATION
Mobile IP
Your scenario combines meaningful node mobility with some infrastructure availability. Mobile IP provides the strongest balance of session continuity, manageable routing, and reliable handoffs without requiring full operator coverage.
Decision confidence
HIGH


Wireless Network Decision Dashboard
Share
Medium (20-100)
Routing complexity
Acceptable routing management overhead
Moderate
Energy constraints
Importance of conserving device power
Moderate
Reliability requirement
Required connection consistency
High
Analyze scenario →
Reset
RECOMMENDED APPROACH
PRIMARY RECOMMENDATION
Mobile IP
Your scenario combines meaningful node mobility with some infrastructure availability. Mobile IP provides the strongest balance of session continuity, manageable routing, and reliable handoffs without requiring full operator coverage.
Decision confidence
HIGH
Why this recommendation?
Parameter-by-parameter decision trace
IMPACT
Infrastructure availability
Limited
Partial infrastructure supports Mobile IP handoff while preserving flexibility.
STRONG
Node mobility
High
Frequent movement strongly benefits from Mobile IP session continuity.
STRONG
Network size
Medium (20-100)
This scale benefits from structured mobility without heavy operator dependence.
MEDIUM
Routing complexity
Moderate
Moderate complexity allows mobility-aware routing without excessive overhead.
MEDIUM
Energy constraints
Moderate
Balanced power needs favor efficient managed handoffs.
MEDIUM
Reliability requirement
High
High reliability favors stable handoff and managed connectivity.
HIGH
NOTE ON METHODOLOGY
Scores are educational estimates based on transparent heuristic rules—not measured performance benchmarks. Real deployments should also evaluate coverage, cost, security, and field-tested performance.
NETSELECT · WIRELESS NETWORK DECISION DASHBOARD
Designed by J. Sai Dheeraj
BL4 · ANALYZE